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Adherence to the nutritional recommendations according to diabetes status in Korean adults: a cross-sectional study

Jong Han Choi¹, Chen Lulu², Seon-Joo Park^{2,3*†} and Hae-Jeung Lee^{2,3,4,5*†}

Abstract

Background Nutritional therapy plays a crucial role in diabetes management. Assessing adherence to nutritional recommendations is critical for evaluating whether the current status of nutrition education is appropriate. This study aimed to evaluate adherence to nutritional recommendations according to diabetes status using data from the Korean National Health and Nutrition Examination Survey from 2016 to 2019.

Methods A total of 2,793 participants (55.4% male) were evaluated regarding their adherence to nutritional recommendations. The “aware” group comprised people who had been clinically diagnosed by a physician. The “treated” group comprised people receiving anti-diabetic medications. The “control” group comprised people who achieved an HbA1c level < 6.5%. The “educated” group comprised people who had received nutrition education or counseling at various locations.

Results Among the 1,918 individuals in the “aware” group, only 243 (8.7%) had received nutrition education. Adherence to nutritional recommendations was generally low, with people with diabetes showing slightly higher adherence to total energy intake (59.6% vs. 55.3%) and total sugar intake (88.0% vs. 84.5%) than people without diabetes. However, adherence to total carbohydrate intake was poor in both the treated and educated groups (34.3% and 26.0%, respectively) compared to the untreated and non-educated groups (44.4% and 36.0%, respectively).

Conclusions These findings indicate inadequate nutritional management for people with diabetes in Korea. Nutrition education should be effectively strengthened to achieve nutritional goals.

Keywords Diabetes mellitus, Nutrition therapy, Guideline adherence

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Background

Nutrition therapy plays a pivotal role in the comprehensive management of diabetes, including glycemic control and prevention of complications [1]. Nutrition therapy aims to induce and maintain nutritionally healthy behaviors through effective education. Most clinical practice guidelines for diabetes management recommend structured education for medical nutrition therapy (MNT) by experienced registered dietitians [1, 2]. However, there are few institutions where MNT by experts is available at a relatively high cost. Therefore, nutritional education is conducted not only in government agencies but also in public health and community centers.

Structured nutrition education delivered by experienced dietitians has been shown to improve dietary habits and health outcomes in people with diabetes [3]. This highlights the importance of enhancing overall nutrition literacy among people with diabetes to ensure that they receive accurate and actionable dietary advice [4]. Studies have indicated that higher nutrition literacy is associated with better dietary adherence and glycemic control [5, 6]. Therefore, reinforcing nutrition literacy through comprehensive education programs can address the barriers faced by people with diabetes in adhering to nutritional recommendations. Despite these efforts in nutrition education and recent advances in novel antidiabetic medications, the proportion of patients who achieve their glycemic control goals has not increased significantly [7, 8]. The low effectiveness of current approaches necessitates an evaluation of how well these educational methods adhere to the recommendations.

The primary objective of our research was to determine the obstacles that impede adherence to nutritional recommendations in Korean adults, both with and without diabetes. To accomplish this objective, we analyzed data from the Korean National Health and Nutrition Examination Survey (KNHANES) and compared adherence across various subgroups classified according to diabetes status (awareness, treatment, control, and education). Through this examination, we aimed to identify critical areas where nutrition education needs to be intensified.

Methods

Study design and participants

The KNHANES has been a national surveillance system conducted by the Korea Centers for Disease Control and Prevention since 1998. This cross-sectional survey assesses the health and nutritional status of approximately 10,000 Koreans annually, collecting data on socioeconomic status, health behaviors, quality of life, healthcare utilization, anthropometrics, biochemical profiles, and dietary intake. The survey included health interviews and examinations conducted by trained staff

at mobile centers, as well as home visits by dietitians. KNHANES provides open access to data for research purposes, supports over 500 publications, and offers annual workshops for data users [9]. For this study, we used data from the 2016 to 2019 surveys, focusing on dietary intake and health-related behaviors. Participants were selected using a multistage stratified cluster sampling method to ensure national representation. Among the 32,379 participants, those with missing information on sampling weight ($n=5,442$), participants aged <30 years ($n=7,838$), pregnant or lactating women ($n=216$), those who reported extreme total energy intake of <500 kcal ($n=181$) or $\geq 5,000$ kcal ($n=128$), and those with missing information on diabetes diagnosis ($n=1,089$) were excluded (Fig. 1). Ultimately, 17,485 participants (male, $n=7,375$; female, $n=10,110$) were included in the final analysis. Individuals with diabetes were defined as participants with an 8-h fasting plasma glucose level ≥ 126 mg/dL or HbA1c $\geq 6.5\%$, those diagnosed with diabetes by a physician, or those treated with antidiabetic medications. According to the given definition, 2,793 individuals (comprising 13.9% of the population) were classified as having diabetes. The nature of the KHANES dataset renders it difficult to determine the specific type of diabetes for each participant with certainty. Nevertheless, previous epidemiological research findings suggest that a large proportion of individuals with diabetes in Korea (approximately 95%) have type 2 diabetes [10]. This suggests that the majority of participants in the present study were also likely to have type 2 diabetes. All participants provided written consent, and the study was approved by the Korean Diseases Control and Prevention Agency Institutional Review Board.

Definition of groups categorized in studies by the status of diabetes

This study included the definition of four groups to categorize people with diabetes. To evaluate adherence to nutrition recommendations in various groups, we defined the following groups as “aware” of diabetes, “treated” with antidiabetic medication, within glycemic goals, “control,” and “educated” about nutrition, and a control group for each. The “aware” group comprised people who had been clinically diagnosed by a physician and the otherwise “unaware” group. The “treated” group included those who were being treated with antidiabetic medications, otherwise categorized as “untreated.” The “control” group was composed of individuals who attained an HbA1c level of less than 6.5%, while those who achieved an HbA1c level of 6.5% or higher were classified as the “uncontrol” group. The “educated” group comprised people who had received nutrition education or counseling at various locations, such as public health centers, district

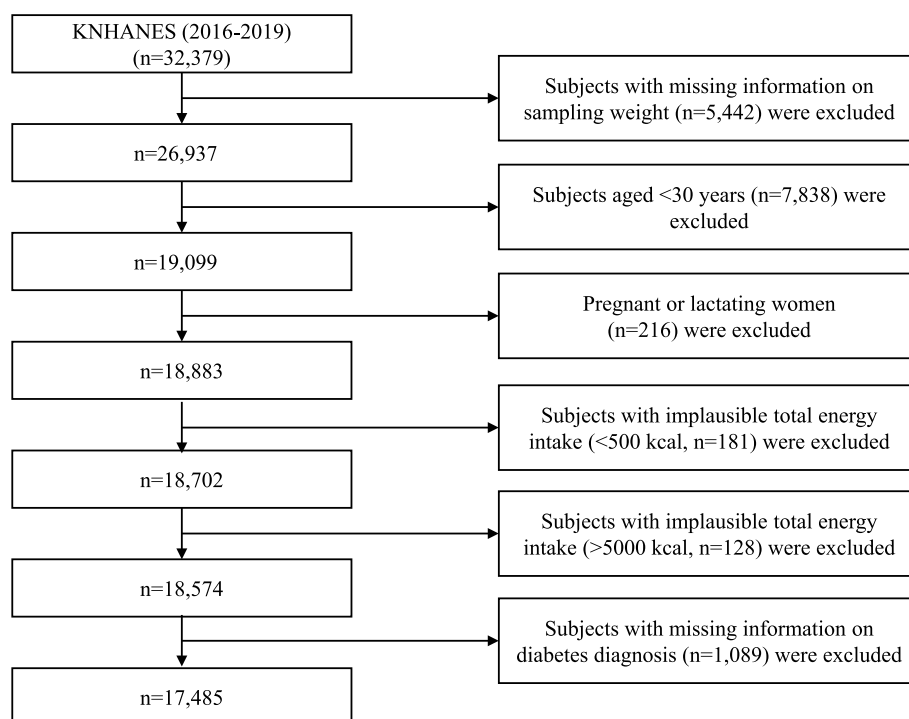


Fig. 1 Flow chart for the participants of the study

offices, community centers, welfare facilities, schools, and hospitals. The last three categorizations (“treated” vs “untreated,” “control” vs “uncontrol” and “educated” vs “non-educated”) are included in the “aware” group, which is based on the consideration of researchers that unaware individuals may lack the motivation to adhere to medication, imply lifestyle management for glycemic control, or receive diabetes-specific nutrition education. This categorization of diabetes status in the KNHANES dataset utilizes the operational definitions used in the Diabetes Fact Sheet of the Korean Diabetes Association (KDA) [7].

Nutritional recommendations for the management of diabetes to evaluate adherence

Nutritional recommendations were based on the 2023 clinical practice guidelines of the KDA [2]. The components were modified based on these guidelines and the 2020 Dietary Reference Intake for Koreans by the Ministry of Health and Welfare and the Korean Nutrition Society. The criteria for evaluating adherence to the nutritional recommendations are as follows: for individuals classified as overweight or obese ($\text{BMI} \geq 23 \text{ kg/m}^2$), the total energy intake should be less than 100% of the estimated energy requirement (EER). For those with a BMI below 23 kg/m^2 , the total energy intake should range between 75 and 125% of the EER. Carbohydrate intake should constitute no more than 65% of the total

energy intake, whereas sugar intake should be limited to 20% or less of the total energy intake. Dietary fiber recommendations are set at a minimum of 30 g per day for men aged 30–64 years, 25 g per day for men aged 65 years and older, and 20 g per day for women aged 30 years and older. Saturated fatty acid intake should be restricted to 7% or less of the total energy intake. Sodium intake should not exceed 2300 mg per day. Alcohol consumption was defined as recommended if there had been no intake in the past year (Table 1).

Variables and measurements

General information, including health-related lifestyle, was collected using a self-report questionnaire. A trained technician collected the anthropometric data. Among the variables, age, sex, BMI (kg/m^2), regular aerobic exercise (yes or no), and alcohol drinking status (not heavy or heavy drinking) were collected. To assess adherence to nutritional recommendations in each group, as categorized according to the diabetes status previously described, daily intakes of total energy (kcal), total carbohydrate (kcal), total sugar (kcal), dietary fiber (g), saturated fatty acid (kcal), and sodium (mg) were also used. Dietary intake of participants of KNHANES was assessed by trained nutritionist using a 24-h dietary recall. Nutrients intakes were calculated and released by KNHAES team.

Table 1 Components for evaluation of adherence to nutritional recommendations for the management of diabetes

Components	Recommendations
1. Nutrition education	Receiving individualized education for medical nutrition therapy (Whether the nutrition education received by the participants in this survey was suitable education on medical nutrition therapy for the management of diabetes was unknown.)
2. Total energy intake	Reducing energy intake for body weight loss in adults with overweight or obesity Overweight or obesity ($\text{BMI} \geq 23 \text{ kg/m}^2$): Total energy intake $< 100\%$ of EER Others ($\text{BMI} < 23 \text{ kg/m}^2$): $75 \sim 125\%$ of EER
3. Total carbohydrate	Reducing total energy intake from carbohydrate Total carbohydrate intake $\leq 65\%$ of total energy intake
4. Total sugar	Minimizing total energy intake from sugar Total sugar intake $\leq 20\%$ of total energy intake
6. Dietary fiber	Sufficient dietary fiber intake Male (30 ~ 64 years old): dietary fiber intake $\geq 30 \text{ g/day}$ Male (≥ 65 years old): dietary fiber intake $\geq 25 \text{ g/day}$ Female (≥ 30 years old): dietary fiber intake $\geq 20 \text{ g/day}$
7. Saturated fat	Minimizing total energy intake from saturated fatty acid Saturated fatty acid $\leq 7\%$ of total energy intake
8. Sodium	Reducing sodium intake Sodium intake $\leq 2300 \text{ mg/day}$
9. Alcohol	Avoiding alcohol consumption if possible No alcohol consumption in a past year

BMI body mass index, *EER* estimated energy requirement

Statistical analysis

We used PROC SURVEYFREQ to calculate the proportion of participants adhering to nutritional recommendations and PROC SURVEYREG to analyze differences between groups. All analyses accounted for the complex survey design of the KNHANES, adjusting for sample weights, strata, and clusters. Statistical significance was set at $p < 0.05$. The differences in general health-related factors and adherence to nutritional recommendations by the KDA guidelines were analyzed between the two groups divided by the status of presence, awareness, pharmacologic treatment, glycemic control, and nutritional education for diabetes. All analyses were adjusted for KNHANES sample weights, strata, and clusters to consider the complex survey design. Statistical analyses were performed using SAS (Version 9.4; SAS Institute, Cary, NC, USA), and a p -value < 0.05 was considered statistically significant after adjusting for age and sex.

Results

Demographic characteristics and health-related lifestyle by diabetes status of adults aged 30+ years in Korea

Table 2 shows the demographic characteristics and health-related lifestyles of the participants according to their diabetes status. Only 65.3% of people with diabetes were diagnosed by a physician as having diabetes, categorizing them in the “aware” group. Among people in the “aware” group, 91.6% received medications; however, only 27.9% achieved the glycemic target of

HbA1c $< 6.5\%$ according to the KDA treatment guidelines [2]. The proportion of people receiving nutritional education in the facilities was extremely low (8.7%). People with diabetes were significantly older (61.5 vs. 51.1 years) and predominantly men (55.4% vs. 48.7%) than those without diabetes. People with diabetes showed a higher BMI than those without diabetes (25.6 vs. 23.8 kg/m^2), although the BMI was lower in the aware group than the unaware group (25.2 vs. 26.4 kg/m^2). In terms of health-related lifestyle, people with diabetes showed lower rates of regular aerobic exercise than those without diabetes (37.4% vs. 44.6%), as well as a higher rate of heavy alcohol drinking (13.7% vs. 11.9%); however, the rate of heavy alcohol drinking was lower in the aware group (11.8% vs. 17.2%). There were no statistically significant differences when the values were adjusted for age and sex (data not shown).

Current status of adherence to nutritional recommendations by diabetes status

Table 3 shows the adherence to nutritional recommendations for each component according to the diabetes status after adjusting for sex and age. Overall, adherence to nutritional recommendations was poor, especially in terms of total carbohydrate, dietary fiber, sodium intake, and alcohol consumption ($< 50\%$). Additionally, awareness of diabetes did not show a significant difference in adherence to most components of the recommendations, except for dietary fiber intake (unaware 45.4% vs.

Table 2 Demographic characteristics and health-related lifestyle by diabetes status of adults 30+ years old from KNHANES 2016–2019

Diabetes										
Diabetes total			Unaware		Aware					
							Unaware	Aware		
									Uncontrolled	Controlled

Table 3 Current status of adherence to the nutritional recommendation according to the status of diabetes (age and sex-adjusted)

	Diabetes									
	Unaware					Aware				
	Diabetes total	Unaware	Aware	Aware total	p	Untreated	Treated	p	Uncontrol	Control
No. of subjects (%)	2,793 (13.9)	875 (34.7)	1,918 (65.3)	1,918 (65.3)		143 (8.4)	1,775 (91.6)		1,363 (72.1)	555 (27.9)
Energy (kcal/day)										
Recommended (%)	59.6	60.4	59.4	59.4	0.7	58.9	59.3	0.94	58.4	61.7
Not recommended (%)	40.4	39.6	40.6	40.6		41.1	40.7		41.6	38.3
Total carbohydrate										
Recommended (< 65%kcal) (%)	47.5	40.1	36.9	36.9	0.20	44.4	34.3	0.03	34.9	35.7
Not recommended (≥ 65%kcal) (%)	52.5	59.9	63.1	63.1		55.6	65.7		65.1	64.3
Total sugar										
Recommended (≤ 20%kcal) (%)	88.0	89.3	88.2	88.2	0.48	87.4	88.3	0.79	88.3	88.1
Not recommended (> 20%kcal) (%)	12.0	10.7	11.8	11.8		12.6	11.7		11.7	11.9
Dietary fiber										
Recommended (%)	47.1	45.4	50.8	50.8	0.02	38.9	50.6	0.01	50.8	52.0
Not recommended (%)	52.9	54.6	49.2	49.2		61.1	49.4		49.2	48.0
Saturated fatty acid										
Recommended (≤ 7%kcal) (%)	72.2	77.0	80.5	80.5	0.10	81.5	81.5	1.00	81.7	80.9
Not recommended (> 7%kcal) (%)	27.8	23.0	19.5	19.5		18.5	18.5		18.3	19.1
Sodium										
Recommended (≤ 2,300 mg/day) (%)	35.5	38.2	39.0	39.0	0.71	38.0	41.0	0.48	40.6	41.2
Not recommended (> 2,300 mg/day) (%)	64.5	61.8	61.0	61.0		62.0	59.0		59.4	58.8
Drinking for a past year										
Recommended (No) (%)	18.3	20.2	21.4	21.4	0.51	21.9	22.1	0.97	22.1	22.2
Not recommended (Yes) (%)	81.7	78.8	78.6	78.6		78.1	77.9		77.9	77.8

BMI body mass index

aware 50.8%, $p=0.02$). Adherence to dietary fiber intake was greater (untreated 38.9% vs. treated 50.6%, $p=0.01$), while adherence to total carbohydrate intake was lower (untreated 44.4% vs. treated 34.3%, $p=0.03$) in the treatment group. Control of diabetes was not associated with adherence to nutritional recommendations. Adherence to total carbohydrate intake was significantly lower in the educated group (non-educated 36.0% vs. educated 26.0%, $p=0.02$). There were no apparent differences in the extent to which recommendations for other components were followed, which could be statistically determined.

Discussion

To the best of our knowledge, this study is the first to evaluate adherence to nutritional recommendations in two groups divided by diabetes status using a nationally representative sample from a nutrition survey in Korea. Of the 1,918 people who were aware of their diabetes, only 243 (8.7%) had received nutrition education. Generally, adherence to nutritional recommendations was low, with only a few groups exceeding 80%. The adherence to recommendations for total energy intake was slightly higher in the diabetes group (59.6%) than in the non-diabetes group (55.3%). Similarly, the diabetes group had a higher adherence to recommendations for total sugar intake (88.0%) than the non-diabetes group (84.5%). Notably, adherence to dietary fiber intake was significantly higher in both the aware and treated groups (50.8% and 50.6%, respectively) than in the unaware and untreated groups (45.4% and 38.9%, respectively). In contrast, adherence to total carbohydrate intake was poor in both the treated and educated groups (34.3% and 26.0%, respectively) compared to the untreated and non-educated groups (44.4% and 36.0%, respectively). These results emphasize the need to improve nutrition education and management in specific areas. These findings are consistent with those of a recent study by McClure et al. using NHANES data in the United States [11], which did not find significant differences in the intake of total energy, macronutrients, and micronutrients according to the presence and control of diabetes.

Our findings revealed a noteworthy pattern of increased carbohydrate consumption among participants in the treatment and education groups. Typical Korean diets have high-carbohydrate, high-fiber, low-fat, and high-sodium dietary patterns centered on steamed rice and salty fermented vegetables [12]. Kim et al. analyzed KHANES data and showed that an unhealthy plant diet, as typical in the Korean diet, increased the risk of metabolic syndrome [13]. Jung et al. recently showed that greater adherence to an unhealthy plant-based diet may increase the risk of abdominal obesity in Korean adults [14]. Additionally,

the KDA announced a new consensus statement to reduce carbohydrate intake and improve glycemic control in individuals with type 2 diabetes [15]. Both the treated and untreated groups, as well as the educated and non-educated groups, would have predominantly consumed a typical Korean diet. Similar dietary patterns across these groups highlight that the typical dietary habits of Koreans may override specific dietary education efforts, suggesting the need for more tailored and impactful nutritional education strategies. Therefore, nutritional education should be strengthened to emphasize the reduction of carbohydrates and healthy plant diets, including whole grains, fresh vegetables, and dairy products.

The issue of concern is the small proportion of participants in our study who had received nutrition education or counseling from facilities such as public health centers, district offices, community centers, welfare facilities, schools, and hospitals. This suggests that a much smaller number of individuals receive personalized education for MNT, as recommended by the diabetes management guidelines [1, 2]. The lack of nutrition education may indicate limited knowledge and understanding of the importance of nutrition in effectively managing diabetes. Inadequate nutrition literacy can serve as an obstacle to following nutritional recommendations, as individuals may encounter difficulties in comprehending dietary guidelines or recognizing the significance of specific dietary elements in managing diabetes [16]. Therefore, enhancing nutritional recommendations through targeted education programs can help bridge these gaps and improve nutrition literacy [16]. Additionally, it is essential to extend support for enhancing individualized education for MNT by experienced dietitians, who have been shown to provide cost-effective care for improving glycemia [17].

This study had a few limitations. First, this was a cross-sectional, observational study using data from the KNHANES. Therefore, our results do not show a causal relationship between diabetes or educational status and adherence to nutritional recommendations. Second, it may have underestimated the number of people who were unaware of their diabetes and had not previously been diagnosed with diabetes by a physician because fasting blood glucose and HbA1c were measured only to evaluate glycemic control in the KNHANES. Lastly, although 24-h dietary recall is a widely used survey method in food intake surveys, it still has limitations in assessing actual nutritional adherence [18]. Therefore, clinical trials are needed to evaluate the quality of nutritional education at each level, using more accurate definitions of diabetes and precise tools for assessing nutritional status.

Conclusion

Our results show that the current nutritional management of people with diabetes in Korea is inadequate to achieve the goals of diabetes management. Therefore, the quality of nutrition education at various levels should be improved and the proportion of people receiving professional nutrition education should be increased.

Abbreviations

BMI	Body mass index
KDA	Korean Diabetes Association
KNHANES	Korean National Health and Nutrition Examination Survey
MNT	Medical nutrition therapy

Authors' contributions

Supervised, conceived and designed the analysis by H-JL and S-JP; collected and analyzed data by CL; Interpreted data and wrote the paper by JHC and H-JL.

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None.

Availability of data and materials

The datasets generated and/or analyzed during the current study are available in <https://www.kdca.go.kr/yhs/yhshmpg/result/yhsresult/rawDtaList.do>.

Declarations

Ethics approval and consent to participate

All participants provided written consent, and the study was approved by the Korean Diseases Control and Prevention Agency Institutional Review Board.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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